

EMS_2018 HOMEWORK 1

1. Three instruments each make 10 repeated measurements of a flow rate known to be $1.500 \text{ m}^3/\text{s}$ with the results given in the next Table.

Trial	Instrument A	Instrument B	Instrument C
1	1.5	1.73	1.552
2	1.3	1.73	1.531
3	1.4	1.73	1.497
4	1.6	1.73	1.491
5	1.3	1.73	1.500
6	1.7	1.73	1.550
7	1.5	1.73	1.456
8	1.7	1.73	1.469
9	1.6	1.73	1.503
10	1.5	1.73	1.493

- Which instrument is most precise? Least precise? Explain.
- Which instrument has the best accuracy? Worst accuracy? Explain.
- Which instrument has the best resolution? Worst resolution? Explain.

2. Draw the **normalized histograms** (the bin width is 1) and calculate the **mean** and the **standard deviation** values for the next data sets:

Boys									
Shoe size	38	39	40	41	42	43	44	45	46
How many	1	2	3	8	25	16	12	9	5

3. Considering the Standard Normal Distribution (z) and using the tables, find:

- $P(z < 1.47)$
 $P(z > -0.25)$
 $P(-2 < z < -1)$
 $P(-1.96 < z < 1.96)$

4. Considering a population of capacitors having a nominal value of $10\mu\text{F}$ and a manufacturing tolerance of 15%. Statistical analysis of the capacitors values suggests that their distribution is normal, having a mean of $\mu=10.02\mu\text{F}$.

- Find the relative error [%] (bias) of the mean value relative to the nominal value (reference value)
- Find the standard deviation knowing that the tolerance is given as $\pm 3\sigma$.
- How many capacitors (from 1000) have values in the interval $\mu \pm 3\sigma$?
- How many capacitors (from 1000) have values in an interval of $\pm 5\%$?
- How many capacitors (from 1000) have values in an interval of $[0\%, 5\%]$?
- How many capacitors (from 1000) have values in an interval of $[-2\%, 4\%]$?
- Find the symmetrical interval (x_1, x_2) where 70% of the population is situated.

Hint: The areas from $\mu \pm 1\sigma$; $\mu \pm 2\sigma$; $\mu \pm 3\sigma$ are known (see the slides from the course lectures).

5. A population of resistances values has the standard deviation of 0.156 ohms. A sample of 60 resistors has the average resistance of 0.55 ohms.

- Based on these data, what is the confidence interval at a 95% confidence level for the population mean resistance.